r/homelab • 9mo ago
Decent-Law-9565

Can the Mellanox ConnectX-4 QSFP28 (MCX455A-ECAT) actually do 100 Gbps?

I read the specification, and on paper it should do up to 100 (if Nvidia isn't lying). However, has anyone actually achieved this? I don't have the resources to actually test it myself so I'd like to know if anyone else has.

0 19

DrDeke • 9mo ago

I have never known Nvidia (née Mellanox) to lie on the datasheet of any of their NICs/HCAs. The MCX455A-ECAT will do 100 gigabit Ethernet or 100 gigabit (EDR) InfiniBand, as per its datasheet.

8

SlothCroissant • 9mo ago

Yeah, no reason to believe Mellanox would have lied about anything on their datasheets - that's just bad business.

FWIW also, if anyone's untrusting of Nvidia for some reason, Mellanox was bought by Nvidia several years after the CX4 series came out. And Mellanox has always been excellent, long before Nvidia bought em.

5

DrDeke • 9mo ago

I miss old school Mellanox. Maybe they were just taking pity on us and our miniscule budget where I worked at the time, but they really went far out of their way (and far beyond what they were contractually obligated to do) to help us on several different occasions.

1

Decent-Law-9565 OP • 9mo ago

Sweet_Dingo_7943 • 9mo ago

For sure, but bypassing the Linux kernel network stack. You'll need to test with tools like `ib_write_bw` over RDMA.

3

skreak • 9mo ago

Yes it will. I have extensive experience with Mellanox chips for years at work. But you'll never see 100gbps with a single transfer between 2 devices using a basic protocol like NFS without extensive tuning. Speeds like that require using RDMA (when in infiniband mode) or RoCE when in ethernet (Rdma Over Coverged Ethernet), sometimes special drivers or filesystems as well. We use 100gbps infiniband

Decent-Law-9565 OP • 9mo ago

I see, this makes sense. I thought I could split that into 4x25 SFP28s but it seems that might not be possible.

1

skreak • 9mo ago

To the best of my knowledge, no. Mellanox switches can do 2 and 4 way splitters for say 4x25 from a single 100gbps port, but ive not come across an option on their HCAs to create 2 or 4 logical interfaces from 1. But, that doesnt mean its not possible, just not something ive come across.

1

1 more reply

zer00eyz • 9mo ago

Yes,

QSFP28 is more than a decade old at this point.

Great article that has a chart with the history of networking modules: https://semianalysis.com/2023/03/17/google-apollo-the-3-billion-game/

2

rxVegan • 9mo ago

Like others have already said: you either need RDMA or multiple parallel TCP streams to saturate the link. At home I'm using SMB over TCP via 25Gb link and while I can get close to saturating it, it's already approaching the point where RDMA would be required to get the most of it.

Nowadays 100Gb is not even that fast. You can do like up to 800Gb but nobody would be using the regular OS ethernet stack at those rates. It's all RDMA.

1

Royal-Imagination398 • 5mo ago

What 100Gbe would you recommend for a person running passive copper 40Gbe on ASUS Z790 motherboards now? I am a photographer and I have large 100 mp files that I need to transfer from one PC to another. Thank you very much, Mike

1

Decent-Law-9565 OP • 5mo ago

Get one of those Mellanox cards as mentioned in the post

1

mastercoder123 • 5mo ago

40gbe since 100gbe is a pretty big jump in price especially if you want a switch

1

GJensenworth • 17d ago

In my homelab, I have a 4 node proxmox cluster with dual 100G links per machine and my main workstations also have dual 100G. The most expensive component per port has been the NICs.

Switches, optics and fiber are all reasonable, but even a used ConnectX-5 card tends to be $200-$400. A Mikrotik CRS520 switch gives you 16x 100G ports for a bit over $2k, and there are Mikrotik and QNAP switches that do 4x100G plus a bunch of other ports for <1k. Used 100G optics can be had for <$10 each.

Another issue is PCIe bandwidth, particularly on desktop motherboards. I have one Threadripper machine with lots and lots of PCIe lanes, but in a typical “gaming” pc, you often find that if you have a GPU in the first slot, the best you can do in another slot is x8. 3.0x8 tops out at 64Gbps, 4.0x8 will do 128Gbps max, and you need either 4.0x16 or 5.0x8 to saturate a dual 100Gbps card.

1

r/homelab • 3y ago

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ConnectX-4 Lx or ConnectX-5 Lx

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r/homelab • 9mo ago

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r/networking • 2y ago

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1 upvote · 6 comments

r/homelab • 3mo ago

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172 upvotes · 129 comments

r/homelab • 4mo ago

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BC-250 any use for homelab?

17 upvotes · 16 comments

r/homelab • 1mo ago

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r/homelab • 9mo ago

Looking for cheap 2.5G switch with 8 port and vlan support. Don't need fancy managed switch.

11 comments

r/homelab • 3mo ago

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10 upvotes · 40 comments

r/homelab • 4mo ago

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r/homelab • 5mo ago

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Homelab Networking -- 10G

136 upvotes · 101 comments

r/homelab • 6mo ago

My new homelab progress, network is up, servers are coming soon

Request: Router/Mini PC with 2x SFP+ 10G & 1x 2.5G PoE+ (repost)

10 upvotes · 21 comments

r/homelab • 4mo ago

Just got my homelab set up to practice CCNA! Excited to start the journey!

89 upvotes · 7 comments

r/homelab • 6mo ago

Massive upgrades to my homelab/network

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